AMIT RAJ

Backend Engineer (Java & Spring Boot)

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PROFESSIONAL SUMMARY

Performance-driven Backend Engineer with **2+ years of experience** architecting, scaling, and delivering production-ready microservices using **Java 17** and **Spring Boot**. Proficient in designing high-throughput REST APIs, asynchronous execution pipelines, distributed caching, and database optimization. Experienced in microservices security (JWT, RBAC), inter-service communication via WebClient, CI/CD automation with Jenkins, and maintaining 70%+ test coverage.

TECHNICAL SKILLS

Languages & Web: Java (8/11/17), SQL, JavaScript, HTML5/CSS3 (Static Web Development & Responsive UI)
Backend Frameworks: Spring Boot, Spring Framework, Spring MVC, Spring Security, Hibernate, Spring Data JPA, REST APIs, Microservices
Distributed Systems & Concurrency: Apache Kafka, Redis, Asynchronous Processing (ExecutorService, CompletableFuture), Spring WebClient
Databases: MySQL, PostgreSQL, MongoDB
DevOps, Cloud, Workflows & Tools: Docker, Jenkins CI/CD, Git, GitHub, Google Workflows & Workspace Automation (Apps Script), Postman, Swagger, Tomcat 9, Canva / Canvas UI Prototyping
Testing & Quality: JUnit 5, Mockito, MockMvc, Unit Testing, Integration Testing
Monitoring & Security: Kibana, Grafana, Elasticsearch (ELK), Centralized Logging, JWT, OAuth2, RBAC, Agile/Scrum

WORK EXPERIENCE

Anantixia Pvt. Ltd. | *Backend Engineer* **Gurgaon, India**
May 2024 – Present

- Architected and scaled robust REST APIs and backend microservices using **Java 17, Spring Boot, Hibernate, and JPA**, handling over **50k+ daily transactions**.
- Engineered an asynchronous task execution system using **Java ExecutorService and CompletableFuture**, decoupling blocking notification workflows and **decreasing API latency by 38%**.
- Optimized relational database persistence by tuning JPA fetch plans and indexing MySQL/PostgreSQL tables, **reducing database CPU load by 40%** during peak traffic.
- Designed fault-tolerant inter-service communication via **Spring WebClient** with automated retry and exponential backoff, boosting cross-service request reliability to **99.8%**.
- Implemented stateless **JWT authentication and granular Role-Based Access Control (RBAC)** using Spring Security filters to protect sensitive enterprise endpoints.
- Authored automated test suites with **JUnit 5, Mockito, and MockMvc** across controller, service, and data layers, raising code coverage from 45% to **70%+**.
- Investigated production incidents and conducted root-cause analysis (RCA) using **Kibana and ELK centralized logging**, reducing Mean Time to Resolution (MTTR) by **30%**.
- Collaborated with DevOps to automate build, test, and release cycles using **Jenkins CI/CD pipelines**.

PROJECTS

Homework Management System | *Java, Spring Boot, Spring MVC, WebClient, JPA, MySQL, Jenkins* **[GitHub]** | ***Project***

- Developed scalable REST APIs for homework creation, assignment, multi-format submission, tracking, and status lifecycle management.
- Integrated multiple internal microservices using **Spring WebClient** to asynchronously validate user roles, courses, standards, and subjects.
- Implemented asynchronous batch assignment with **Java ExecutorService**, assigning tasks to 5,000+ student groups concurrently without blocking workflows.
- Optimized high-volume JPA/Hibernate queries and transaction boundaries, **reducing database query latency by 35%** under peak load.

Student Management System | *Java, Spring Boot, Spring Security, Hibernate, MySQL, JWT, Jenkins* **[GitHub]** | ***Project***

- Engineered core backend services for student onboarding, academic profile tracking, course enrollments, and sections management for 50k+ records.
- Implemented secure onboarding workflow generating temporary hashed credentials and integrated asynchronous email notifications upon registration.
- Devised a concurrency-safe unique student ID generation algorithm combining timestamp, tenant ID, and machine ID to eliminate race conditions.
- Refactored database schemas and optimized indexing in MySQL, boosting data throughput by **40%** during peak admissions.

Employee Management System | *Java, Spring Boot, Spring Security, REST API, MySQL, JWT, Jenkins* **[GitHub]** | ***Project***

- Developed backend APIs for employee onboarding, departmental role assignments, designations, and team hierarchy management.
- Utilized **Spring Security Context** to extract, validate, and propagate authentication tokens for downstream microservices.
- Enforced fine-grained authorization using Spring Security filters, method-level security, and Role-Based Access Control (RBAC).
- Tuned bulk database operations and connection pools to support seamless data scaling without performance degradation.

EDUCATION

Government Engineering College, Banka **Banka, Bihar**
Bachelor of Technology (B.Tech) ***2019 – 2023 | CGPA: 7.74***